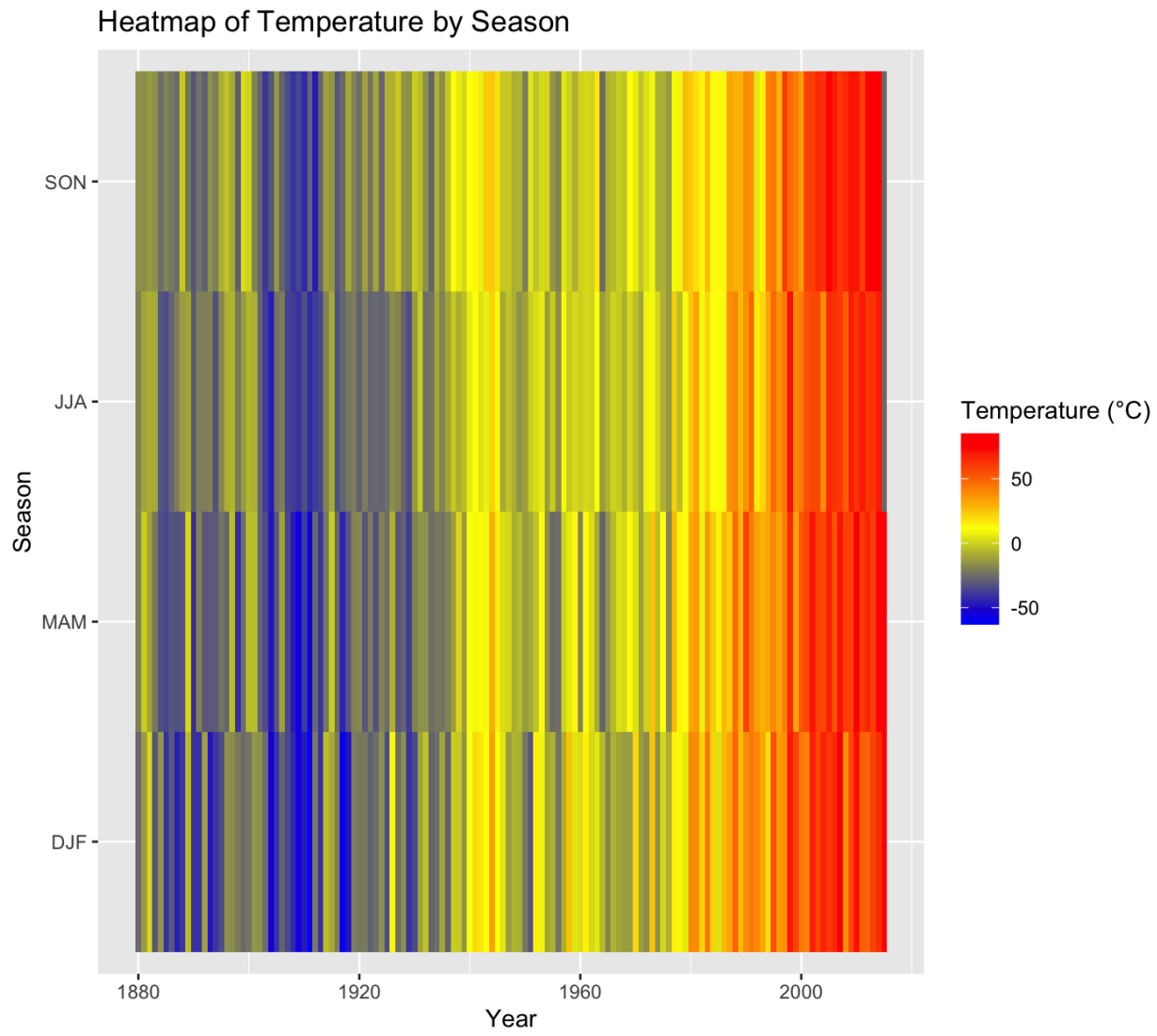
NASA GISS Analysis

Summary

This report is a detailed analysis of worldwide temperature anomalies between 1880-2015. Various data visualization tools—a heatmap, a scatter plot, small multiples, and a box plot—generated via R studio, have been employed to analyze the data.

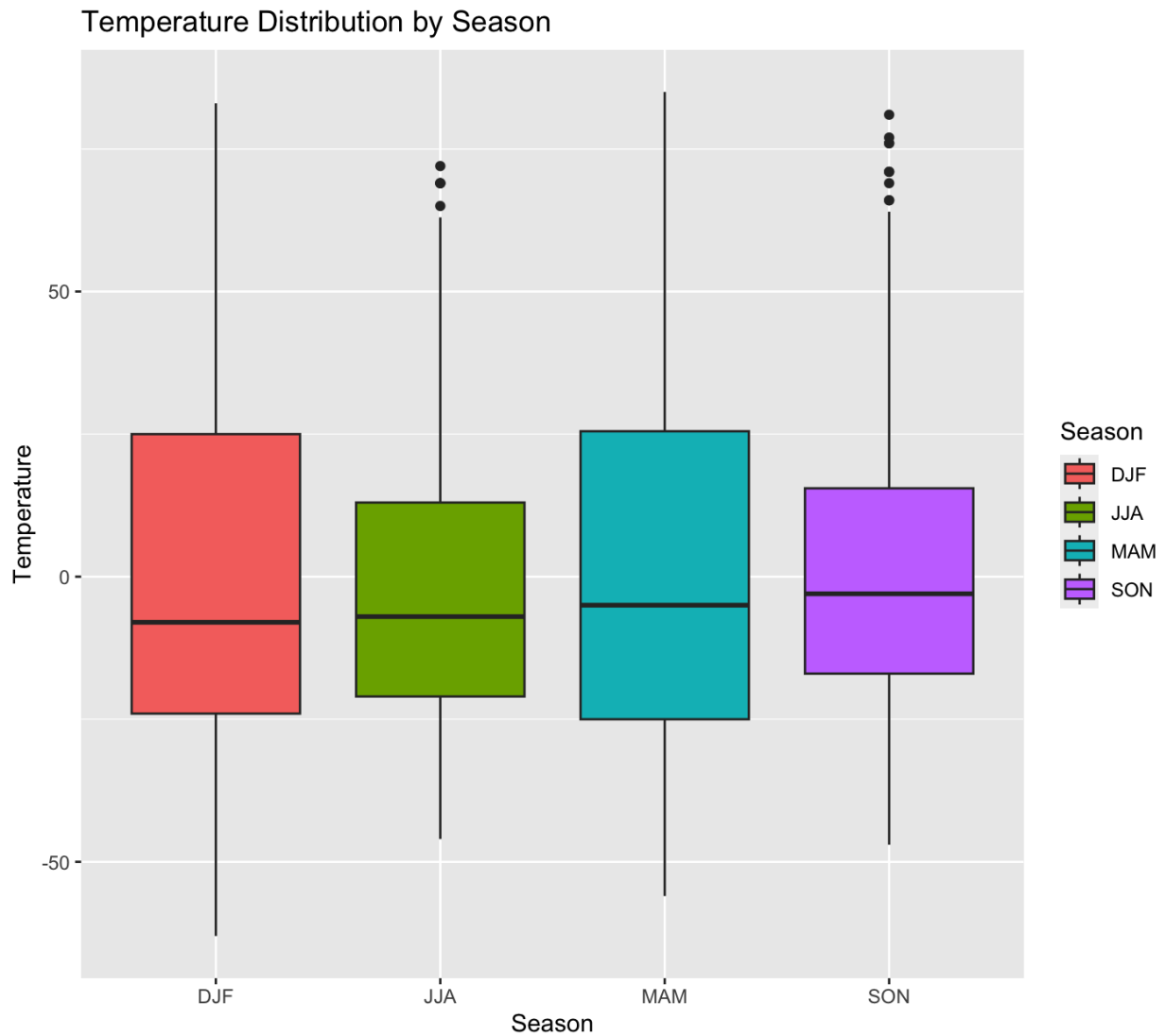
Analysis of Temperature Anomalies Over the Years by Season



The heatmap above depicts temperature changes across seasons from 1880 to 2015. Blue shades represent colder temperatures, with yellow and orange indicating warmer temperatures. The darker the blue, the colder the temperatures, and the darker the orange, the warmer the temperatures. The coldest recorded temperature in the dataset is -126 degrees Celsius, while the warmest is 136 degrees Celsius.

In 1880, Winter, Spring, Summer, and Fall showed varying degrees of cold temperatures, with Fall and Summer exhibiting slightly warmer temperatures compared to Spring and Winter. However, as the years progress, all seasons experience a trend towards warmer temperatures. By the 21st century, the lowest average temperatures are notably warmer than those recorded in 1880, reflecting a consistent overall warming trend across seasons.

Analysis of Temperature Anomaly Distribution by Season



This boxplot illustrates temperature anomalies across the four seasons.

Winter (DJF): The interquartile range and maximum temperatures suggest that most anomalies range from moderately warm to extremely cold. Values between the interquartile range and the fourth quartile indicate moderately warm to extremely warm temperatures.

Spring (MAM): Anomalies range from moderately warm to extremely cold, as indicated by values falling between the interquartile range and the first quartile. Values within the fourth quartile suggest moderately warm to extremely warm temperatures.

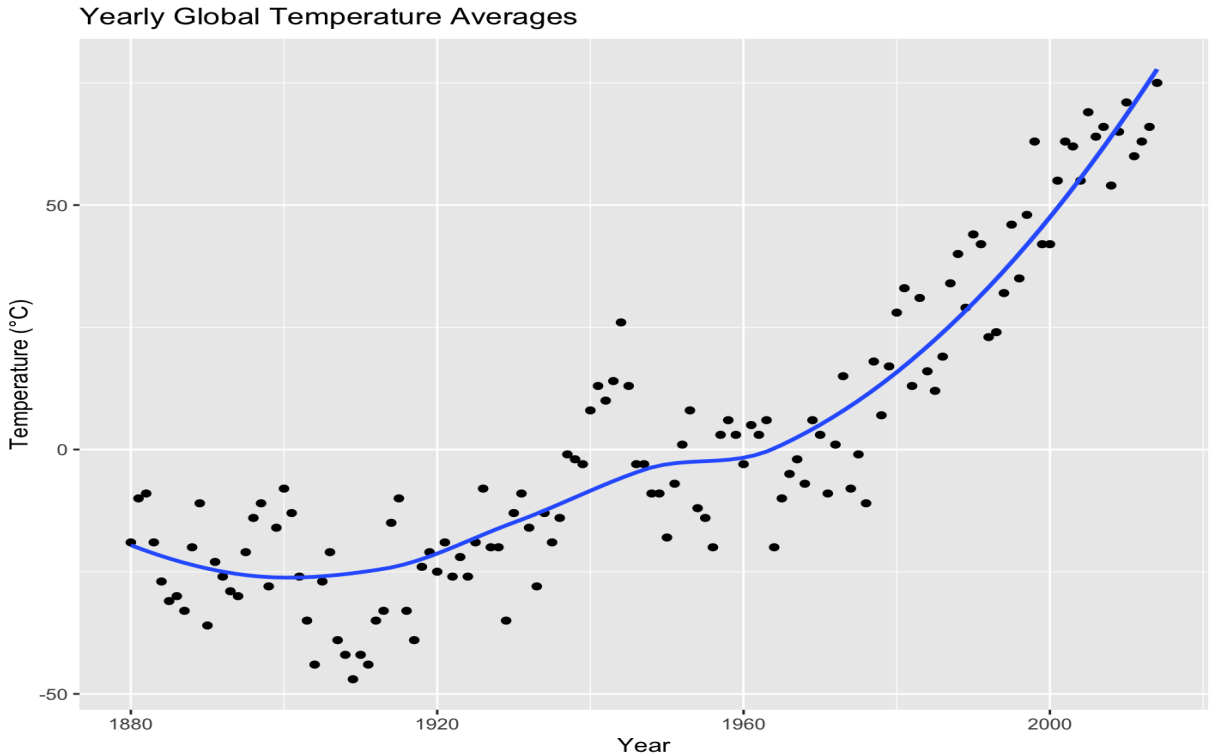
Summer (JJA): Data between the first and third quartiles represent moderately warm to extremely cold temperatures. Anomalies within the fourth quartile indicate moderately warm to extremely warm temperatures, with occasional extremely hot outliers.

Fall (SON): Anomalies between the first and third quartiles range from moderately warm to extremely cold. Temperatures in the first quartile indicate moderately warm to extremely warm temperatures, with occasional extreme heat outliers.

Interestingly, both Winter and Spring exhibit some warmer temperature anomalies than Summer.

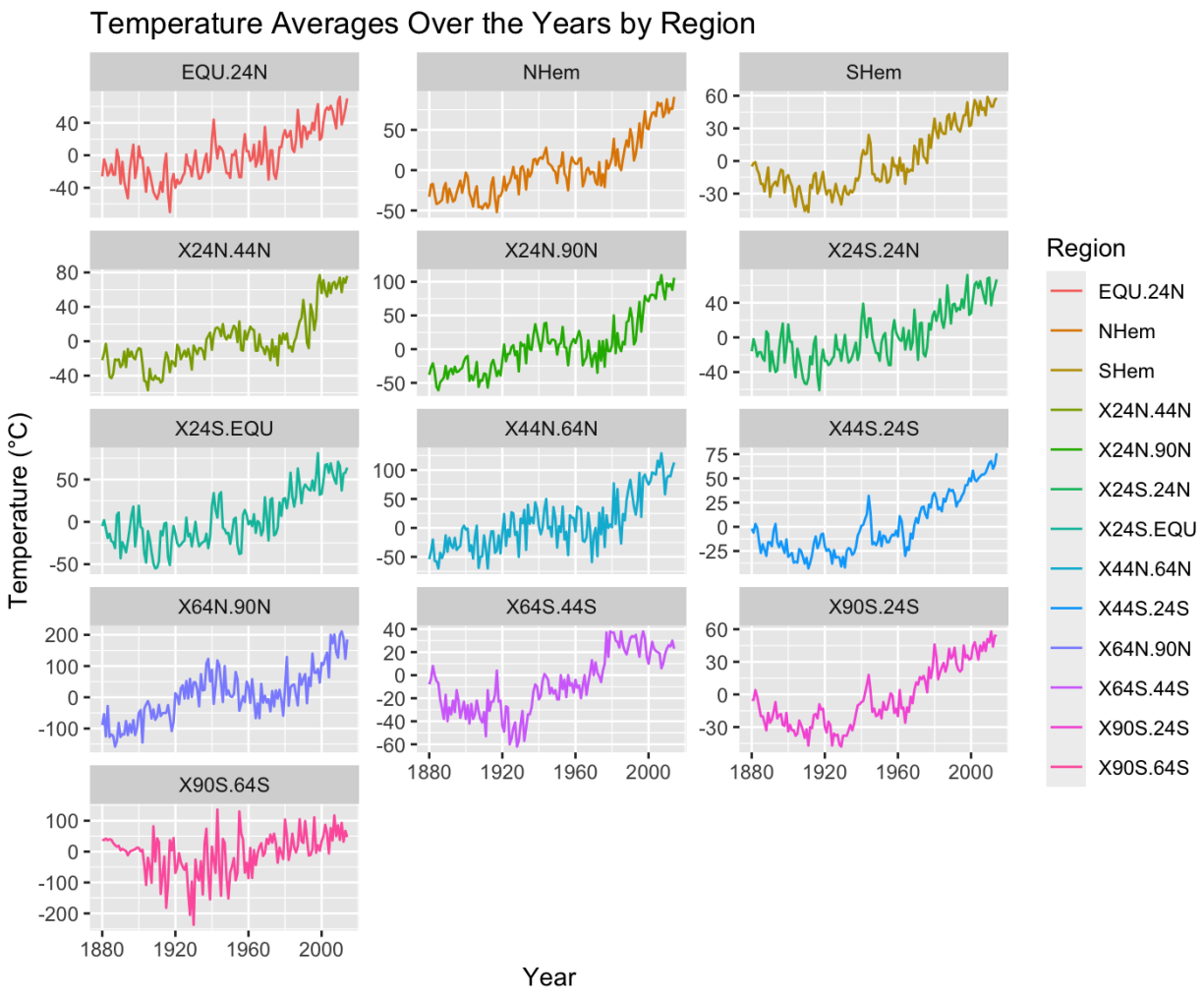
The boxplot illustrates distinct patterns across seasons. Winters typically exhibit a wide range of temperatures, from moderately warm to extremely cold, with occasional warm temperature anomalies. Springs and Falls also exhibit variability, with anomalies ranging from moderately warm to extremely cold, alongside warm outliers. Summers consistently experience warmer anomalies, with temperatures typically ranging from moderately warm to extremely hot, occasionally reaching extreme levels. Overall, the distribution of values in this box plot suggests seasonal variations in temperature anomalies, with summers consistently demonstrating the highest temperatures and winters showcasing a broader range of temperature extremes.

Analysis of Yearly Global Temperature Anomaly Averages



The above scatterplot reveals a clear upward trend in global temperature anomalies over time. As the years progress, there is a noticeable increase in the average global temperatures. Most temperature anomaly values are above 0ºC, indicating that, on average, global temperatures have been getting warmer over the years. Moreover, the rate of temperature increase seems to have accelerated in recent decades, particularly post-1960. However, a few data points deviate significantly from the trend line, representing outliers. These outliers include some extremely warm and some extremely cold temperatures. Pre-1980, most outliers indicate extremely cold temperatures, and post-1980, most outliers fall between moderate to extreme heat. This shift reflects a warming trend in recent decades. It also suggests that while extreme cold events still occurred, they became relatively less frequent compared to extreme heat events, which became more common through 2015.

Analysis of Temperature Anomaly Averages Over the Years by Region



The graph above represents temperature anomalies in different regions from 1880 – 2015.

Region EQU 24N:

The temperature anomaly fluctuates between approximately -50°C and 40°C. There is also a significant increase in positive anomalies after the mid-20th century.

Northern Hemisphere:

The graph indicates a consistent upward trend in the Northern Hemisphere, with temperature anomalies ranging from approximately -80°C to 50°C. The graph also indicates that temperatures in the Northern Hemisphere have generally been increasing over the years, suggesting a continuous warming trend in the region. Additionally, the positive anomaly becomes more pronounced post-1960. This rate of temperature increase shows that the Northern Hemisphere has rapidly warmed up in the decades between 1960 and 2015.

Southern Hemisphere:

The temperature anomaly in the Southern Hemisphere follows a similar pattern to the Northern Hemisphere but with slight differences. The rate of warming seems to be less pronounced in the Southern Hemisphere than it is in the Northern Hemisphere.

X24N-44N & X44N-64N:

Both regions show significant fluctuations in temperatures from 1880-2015.

Overall, both regions exhibit a positive trend in temperature anomalies, which like the Northern and Southern Hemisphere also indicates a warning trend over the years.

X24S-EQU & X44S-24S:

Like the Northern Hemisphere, these regions also exhibit increased positive anomalies over the years despite showing fluctuations in temperature. However, the rising temperatures in both regions are less significant than in the Northern Hemisphere.

X64N-90 & X90S-64S:

There is a range of variability in the temperature anomalies of these regions. The sharp increase in positive temperature anomalies toward the end of the century for region X64N-90 is noteworthy. The graph for this region also exhibits extremely cold temperatures toward the end of the 1800s, compared to the extreme heat of up to 200°C. seen in the 21st century. The graph also shows that sometime around the 1930s, the temperature anomalies continued to increase over time. The extremely cold temperatures seen between 1880 and 1920 never occur again as of 2015.

For region X90S-64S, the graph exhibits extreme to moderately cold temperature anomalies toward the late 1800s. However, between the late 1910s to the late 1920s, there are some significant dips in temperature to lower than -200°C. From the 1960s well into the 20th century, the temperature anomalies exhibit both extreme cold and extreme heat.

Conclusion: The collective trend across all examined regions indicates rising temperature anomalies over time, especially post-1960.